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```
PS C:\Users\msi> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 24H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Restarting existing docker container for "minikube" ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\msi> kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   10m   v1.35.1
```

deployment.yaml | service.yaml

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

deployment.yaml | service.yaml

service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - port: 80
      targetPort: 80
  type: NodePort
```

```
PS C:\Users\msi> cd kubernetes
PS C:\Users\msi\kubernetes> kubectl apply -f deployment.yaml
deployment.apps/web-nginx created
PS C:\Users\msi\kubernetes> kubectl apply -f service.yaml
service/web-nginx-service created
```

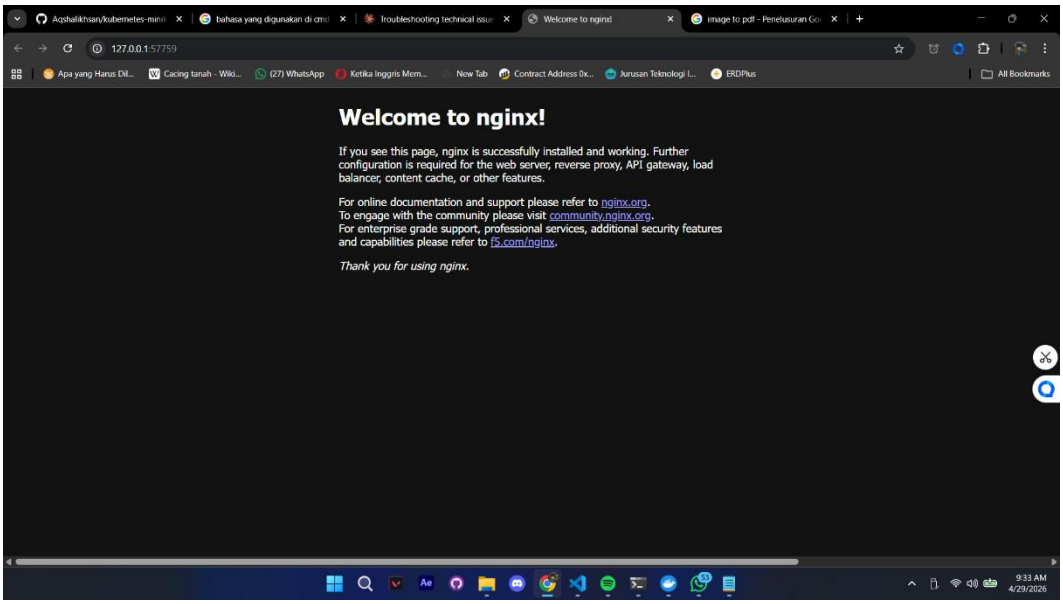
```
PS C:\Users\msi\kubernetes> kubectl get all
NAME                              READY   STATUS              RESTARTS   AGE
pod/web-nginx-748d764f7-6nfbc     0/1     ContainerCreating   0          23s
pod/web-nginx-748d764f7-t6nkn     0/1     ContainerCreating   0          23s
pod/web-nginx-748d764f7-wx5gq     0/1     ContainerCreating   0          23s

NAME                          TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
service/kubernetes            ClusterIP   10.96.0.1       <none>        443/TCP        13m
service/web-nginx-service     NodePort    10.109.33.242   <none>        80:31514/TCP   11s

NAME                          READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx     0/3     3            0           23s

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-748d764f7      3         3         0       23s
PS C:\Users\msi\kubernetes> kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-6nfbc     1/1     Running   0          34s
web-nginx-748d764f7-t6nkn     1/1     Running   0          34s
web-nginx-748d764f7-wx5gq     1/1     Running   0          34s
PS C:\Users\msi\kubernetes> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-6nfbc     1/1     Running   0          43s
web-nginx-748d764f7-t6nkn     1/1     Running   0          43s
web-nginx-748d764f7-wx5gq     1/1     Running   0          43s
```

```
PS C:\Users\msi\kubernetes> minikube service web-nginx-service
|-----------|-----------------|-------------|---------------------------|
| NAMESPACE |      NAME       | TARGET PORT |           URL             |
|-----------|-----------------|-------------|---------------------------|
| default   | web-nginx-service | 80          | http://192.168.49.2:31514 |
|-----------|-----------------|-------------|---------------------------|
* Starting tunnel for service web-nginx-service.
|-----------|-----------------|-------------|-----------------------|
| NAMESPACE |      NAME       | TARGET PORT |          URL          |
|-----------|-----------------|-------------|-----------------------|
| default   | web-nginx-service |             | http://127.0.0.1:57759 |
|-----------|-----------------|-------------|-----------------------|
* Opening service default/web-nginx-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```




```
PS C:\Users\msi\kubernetes> kubectl get pods -w
NAME                        READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-6nfbc   1/1     Running   0          7m25s
web-nginx-748d764f7-t6nkn   1/1     Running   0          7m25s
web-nginx-748d764f7-wx5gq   1/1     Running   0          7m25s
web-nginx-748d764f7-g6xb8   0/1     Pending   0          0s
web-nginx-748d764f7-b29hh   0/1     Pending   0          0s
web-nginx-748d764f7-g6xb8   0/1     Pending   0          0s
web-nginx-748d764f7-b29hh   0/1     Pending   0          0s
web-nginx-748d764f7-g6xb8   0/1     ContainerCreating   0          0s
web-nginx-748d764f7-b29hh   0/1     ContainerCreating   0          0s
web-nginx-748d764f7-g6xb8   1/1     Running             0          3s
web-nginx-748d764f7-b29hh   1/1     Running             0          5s
web-nginx-748d764f7-b29hh   1/1     Terminating         0          81s
web-nginx-748d764f7-g6xb8   1/1     Terminating         0          81s
web-nginx-748d764f7-b29hh   1/1     Terminating         0          81s
```

```
PS C:\Users\msi\kubernetes> kubectl scale deployment web-nginx --replicas=5
deployment.apps/web-nginx scaled
PS C:\Users\msi\kubernetes> kubectl scale deployment web-nginx --replicas=3
deployment.apps/web-nginx scaled
PS C:\Users\msi\kubernetes>
```

```
PS C:\Users\msi\kubernetes> kubectl get deployment -w
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx   3/3     3            3           9m3s
web-nginx   3/3     3            3           13m
```

```
PS C:\Users\msi\kubernetes> minikube node add
😄  Adding node m02 to cluster minikube as [worker]
❗  Cluster was created without any CNI, adding a node to it might cause broken networking.
👍  Starting "minikube-m02" worker node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔥  Creating docker container (CPUs=2, Memory=2200MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
🏄  Successfully added m02 to minikube!
PS C:\Users\msi\kubernetes> kubectl get nodes
NAME           STATUS     ROLES           AGE   VERSION
minikube       Ready      control-plane   23m   v1.35.1
minikube-m02   NotReady   <none>          17s   v1.35.1
```

```
PS C:\Users\msi\kubernetes> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running
```

```
PS C:\Users\msi\kubernetes> kubectl delete -f deployment.yaml
deployment.apps "web-nginx" deleted from default namespace
PS C:\Users\msi\kubernetes> kubectl delete -f service.yaml
service "web-nginx-service" deleted from default namespace
```